

Cassie Mini-Project Approach

Team Name: Gain of Thrones

Top Score: 239.96

Details on our winning submission (controllers / techniques used):

- Our submission employed a force-control strategy with a PD fallback, activated when the optimizer could not find a feasible solution. We also implemented a dynamic downward desired COM motion to enhance rotational stability under dynamic conditions.

Any cool Graphics / Equations:

- Look, we made it take a step! - <https://youtu.be/7VkbnPQIUvo?si=ijMVahk0X2vaYCIT>
- For equations, please see Lecture 15 and Lecture 16 as well as the papers attached below

Other Techniques we tried:

- We also spent significant time developing a step approach
 - Developed an IK solver using `computeFootPosition()`.
 - Implemented a state machine and trajectory planner, inspired by drone lab firmware.
 - Our IK was initially solved in the world frame; we determined that solving in both world and body frames would be necessary.
 - Considered controlled lowering of the center of mass (COM) by solving for the stance foot position relative to the ground, rather than manually lowering it.
 - Attempted to make the robot more dynamic by shifting the stance leg to generate upward push and lateral stepping:
 - The IK calculator already supports desired foot positions below ground, enabling upward motion of the body and leg extension calculations.
 - Calculated the full inverse kinematics in the world frame, then overrode the swing leg solution using relative body-frame kinematics.
 - Observed that the stance leg induced torso pitch, which destabilized the system:
 - Once pitched, the swing leg is resolved relative to the torso → the entire system becomes unstable
- Tried implementing stepping with force control when the main force controller failed to converge.
- Attempted to stabilize the COM anywhere inside the support polygon, rather than forcing it back to the nominal position.
- Experimented with relaxing optimization constraints.
- Adjusted the force controller behavior when not all contact points were grounded.
- Tried algorithmically lowering force-control gains to improve feasibility.
- Ultimately, we achieved a competitive result using the simple PD controller initially provided in the script.

What we learned from this project:

- Practical implementation of force-control for a standing robot.
- How to formulate frictional constraints within a quadratic optimization problem.
- The inefficiency and limitations of `fmincon` for real-time control.
- Multiple approaches to formulating QPs, and how small modeling choices impact performance.
- The importance of correctly understanding state rotations, especially when they are defined in the opposite convention than expected.
- Improved skills in parsing and interpreting notation in academic papers.
- Techniques for sequentially tuning control gains to achieve stability.
- That stepping is extremely difficult - and often overrated compared to other stabilization strategies.

References we found very useful for this project:

- C. Ott, M. A. Roa, and G. Hirzinger, "Posture and balance control for biped robots based on contact force optimization," in *Humanoid Robots (Humanoids), 2011 11th IEEE-RAS International Conference on*. IEEE, 2011, pp. 26–33.
 - The derivations for the grasp matrix and equations of motion were much clearer in this publication than in the following, but even then it took a lot of effort decoding the notation of the paper and comparing it with the helper_functions in the provided zip file.
- B. Henze, M. A. Roa, and C. Ott, "Passivity-based whole-body balancing for torque-controlled humanoid robots in multi-contact scenarios," *The International Journal of Robotics Research*, vol. 35, no. 12, pp. 1522–1543, 2016.
- J. Pratt, J. Carff, S. Drakunov, and A. Goswami, "Capture point: A step toward humanoid push recovery," in *Humanoid Robots, 2006 6th IEEE-RAS International Conference on*. IEEE, 2006, pp. 200–207.
 - Not very application for the step